BTS – VII – 11.15 –0774 Reg. No. [ ] C

B.Tech. Degree VII Semester Examination November 2015

CS 1704 ANALYSIS AND DESIGN OF ALGORITHMS
(2012 Scheme)

Time : 3 Hours Maximum Marks : 100

PART A
(Answer *ALL* questions)

(8 × 5 = 40)

I. (a) What are the different asymptotic notations used to compare different algorithms?

(b) Solve the recurrence equation $T(n) = 2T\left(\sqrt{n}\right) + 1$.

(c) List out the properties of Red Black Tree.

(d) What is Amortized Time Analysis?

(e) Explain graph traversal techniques.

(f) Explain strongly connected component algorithm

(g) Write short note on approximation algorithms.

(h) Explain complexity class NP.

PART B

(4 × 15 = 60)

II. (a) Explain Recursion Tree. (5)

(b) Explain with an example the concept of dynamic programming. (10)

OR

III. Explain the various criteria used for analyzing algorithms with suitable example. (15)

IV. Explain anyone searching algorithm with an example. Also derive the worst case and average case time complexity. (15)

OR

V. Write the merge sort algorithm and analyze the worst case and average case time complexity. (15)

VI. Explain any one algorithm for finding the ALL pair shortest path in graph and analyze its complexity. (15)

OR

VII. What is a binary search tree? Explain an algorithm for constructing an optimal binary search tree. Analyze its complexity. (15)

VIII. Define travelling salesman problem. Explain the two possible strategies for TSP. (15)

OR

IX. What is bin packing problem? Explain the first fit decreasing strategy for solving bin packing problem. (15)

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